SHIMADZU

# PACKING SLIP

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By:SHIMADZU\accellossvc

Shimadzu Scientific Instruments, Inc.
7102 Riverwood Drive, Columbia, MD
21046-2502 USA
(410) 381-1227 Fax - (410)381-1222

**Ship To:** UNIV OF NV-RENO
1664 N Virginia St
RENO, NV 89557-0388
USA

**Ship To Attn:** PO # PO-01-00030474

**Contact:** Fred Harvey

**Phone:** 775-784-6019

**Sales Order:** IN0012102

**Order Date:** 8/31/2023

**Customer ID:** 416001

**PO Number:** PO-01-00030474

| Sales Rep ID | FOB | Freight Terms | Incoterms | Shipping Method | Payment Terms |
|---|---|---|---|---|---|
| 10AXA | DESTINATION | PREPAID & ADD | | UPS | NET 30 DAYS |

| Ordered | Shipped | B/O | Item / Serial/Lot Qty | Description / Serial/Lot Number | U of M |
|---|---|---|---|---|---|
| 1 | 1 | 0 | 224-20300-58 | MALDI-8020 Benchtop MALDI-TOF MS | EA |
| | | | 1 | O30131700168KR | |
| 3 | 3 | 0 | 071-60815-07 | Power Cable for MALDI instruments | EA |
| 1 | 1 | 0 | TO-430R00 | MALDI MS Microscope Slide 48 Well Sample Plates (Fleximass-DS) Polymeric (16 plates) | EA |
| 1 | 1 | 0 | 220-91111-07 | MALDI Welcome Letter | EA |
| 1 | 1 | 0 | TO-724R00 | Reagents, Biotech, MALDI Matrix and Calibrants Kit | EA |
| 1 | 1 | 0 | 220-91649-10 | Barcode Scanner - Life Science | EA |
| 1 | 1 | 0 | TO-431R00 | MALDI MS Microscope Slide 48 Well Sample Plates (Fleximass-SR48) Stainless Steel (4 plates) | EA |
| 1 | 1 | 0 | 1YW | 1 YEAR WARRANTY | EA |
| 1 | 1 | 0 | TIER 2 I&F | Tier 2 Installation and Customer Familiarization | EA |
| 2 | 2 | 0 | 220-97300-01 | MALDI MS Training Course Customer Site/Day | EA |

## CERTIFICATE OF COMPLIANCE

It is hereby certified that all equipment and parts furnished are in conformance with the requiirements, specifications and drawings applicable to your purchase.

THANK YOU FOR YOUR ORDER

## Extremely Important Attention Receiving Dept.

1. Check the shipment and note any shortage or damage on delivery receipts
2. Do not refuse the damaged merchandise
3. Keep carton, packing material, and goods for carrier inspection
4. Call the delivery carrier immediately for any investigation
5. Claims must be made within 5 days after the date of delivery
6. DO NOT return goods without obtaining a return authorization number
7. Call our Columbia Office at 1-800-477-1227 or 410-381-1227

Notice: This product may or may not contain phenol, isopropylated, phosphate (3:1) Per 47 CFR 751.407,The Environmental Protection Agency prohibits processing and distribution of this chemical/product for any use other than: (1) In hydraulic fluids either for the aviation industry or to meet military specifications for safety and performance where no alternative chemical is available that meets U.S. Department of Defense specification requirements, (2) lubricants and greases, (3) new or replacement parts for motor and aerospace vehicles, (4) as an intermediate in the manufacture of cyanoacrylate glue, (5) in specialized engine air filters for locomotive and marine applications, and (6) in adhesives and sealants before January 6, 2025, after which use in adhesives and sealants is prohibited. In addition, all persons are prohibited from releasing PIP (3:1) to water during manufacturing, processing, and distribution in commerce, and must follow all existing regulations and best practices to prevent the release of PIP (3:1) to water during the commercial use of PIP (3:1).

$1.07 \times 10^{-6}$ mbar

$5.0 \times 10^{-6}$ mbar Analysis pressure

Laser power 81

MALDI solution → program
Kratos. application

Username: MaldiSolutions Admin
password: shimadzu

right click go to position

my uses
Smspain
bud Ka$124

pulsed extraction

we only have linear to linear

Laser power range 1 – 180

solid red services 3 Windows not ready

Acquire → operate to ramp voltages

remember plate ? turns

(2)

Calibration?

---

Calibration

other missing parts

Cal. mix

−20°C Freezer

BSA

TOF2 Mix

matrixes

(1) to open data

11/13/2023 ①

Analyzer pressure after weekend $5.94 \times 10^{-7}$

— shimadzu folder
current Instrument Status

Administration Console : Where you make users

11/13/2023 bol
about 40 seconds
to pump down
after target lacs

if you undock laser window
just close it & it will redock

---

Target
1 X 48 2.8 mm wells ( 431 RØØ)

$5 \times 10^{-6}$ mbar

(2)
Sup

Initial set: should only need to do once

Target Tab → Target 1x48 2.8 mm wells [431RØØ]

Acquire Tab → Tuning

Instrument Mode

↳ Linear

↳ Polarity

↳ Positive

↳ Matches

↳ Linear

---

(7) click operate

(8) setup acquisition

H K S S M
9 10 11 12 13

(a) mass range

(b) Spots Manual positioning

☑ Dither

(c) Accumulate 5 shot @ 50 Hz } these go-can change

(d) Profile 25 profiles

(e) pulse extract put in desired mass

(f) Blanking a cutoff (at least 500)

(d) laser power 35 too much will affect peak position & resolution

SOP

1. you're going to logging to the data acquisition software using your NetId
2. all data is stored in the instrument software database under your netid
3. launch "Current Instrument Status"
   - click connect
   - ~~this step~~ launching this is not required but gives nice overview
4. Launch "data acquisition" software
   click next
   put in your log in info
5. to open door
   Acquire Tab → Open Door
   blinking Amber → blinking Green / open door manually
6. Insert plate & close door
   click confirm in software

J. tof-vgr



P. microflex

Subject: MALDI TOFMS and MestReNova

Date: Friday, May 9, 2014 at 12:18:19 PM Pacific Daylight Time

From: Stephen M Spain

To: June Pinky T Acay, Arjun Acharya Acharya, Rameez Ali, Mario A Alpuche, Devendar Anumandla, Alissa C Backman, krishna barakoti, Korry Barnes, Thomas W Bell, Thomas M Brown, Jonathan Buttrick, Rachael A Carmichael, Sean M Casey, Vincent J Catalano, Wesley Chalifoux, Chiraphorn Khan, Kelly Chen, Pushpa Chhetri, Heather Clewett, Travis D Cournoyer, Ana de Bettencourt Dias, Raphel A Enow, Kent M Ervin, Ashantha Fernando, Brian J Frost, Surja B Ghale, Khagendra B Hamal, Brady J Janes, Christopher S Jeffrey, Carey Johnson, Awanthi P Kalahe Hewage, Nelum M Karunathilake, Rezvan Kazemi, Benjamin King, Jerry G Lanorio, Jocelyn A Pineda-Lanorio, Michael D Leonard, Ryan Littlefield, Joshua Zimmerman, Kaitlin McDermott, Bevan G Meade, Daniel J Murray, Trang Nguyen, LingLee Ounkham, Suman Parajuli, Dustin Patterson, Neluni Perera, Jeffrey Rossini, Jennifer Schmitt, Jason Shearer, Robert S Sheridan, Stephen M Spain, William Thompson, Michael Walley, Pei Wang, Radha Bam, Farzaneh Chalyavi, Geeta Chaudhary, Amy Cunningham, Robert Gilley, Manuel A Gracia-Nava, David Hogle, Jade Horton, Khoa Le, Liezel Lumangtad, Drew N Maurine, Andrew J Pohlman, Rodney A Tigaa

If you want to process your MALDI-TOFMS data in MestReNova you'll have to

**Windows:**

One time only, configure MestReNova to read Bruker Compass files

1) uninstall MestReNova
2) install CompassXtract 32-bit version from Bruker's website (the 64-bit version will not work)
3) reinstall MestReNova making sure to select the checkbox for Bruker files when installing

Open the data on your computer

1) when you save data from the MALDI-TOFMS it will create a folder with a lot of stuff in it, the file you want to open is "fid", open the fid file in MestReNova

**Macintosh:**

1) open your data in Flexanalysis on the MALDI-TOFMS computer
2) export it: File → export → raw spectrum as mzXML and save it your directory on argenta
3) in Mestrenova *on your Mac* open the mzXML file your created in step 2

or

1) on the MALDI-TOFMS computer open the data file directly in MestReNova
2) turn off centroid by right-clicking on your spectrum and selecting ***Toggle Centroid Spectrum***
3) save the MestReNova file to argenta
4) go to *your Mac* and open the MestReNova file that you saved to argenta

**vailable Matrices (see Aldrich chart for applications)**

045992 - 3-hydroxypicolinic acid
045994 - 2'-(4-hydroxyphenylazo)benzoic acid
045996 - 2,5-dihydroxybenzoic acid
071916 - sinapic acid
045991 - 3-indoleacrylic acid

045995 - 3,5-dimethoxy-4-hydroxycinnamic acid (sinapic acid)
045985 - trans-4-hydroxy-3-methoxycinnamic acid (ferulic acid)
147638 - trans-2-[3-(4-tert-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB)

## Sample preparation procedure (recommendation)

***Pre-Remarks***

Poor sample preparation will degrade sensitivity, yield low resolution and poor reproducibility. The generation of ions through MALDI depends on the production of a suitable composite material, consisting of the matrix substance and the analyte. For best results use only chemicals of highest available purity.

***Chemicals and materials required***

Acetonitrile (ACN)
0.1% Trifluoroacetic acid (0.1% TFA) in water

: *Acetonitrile is classified as hazardous chemical (DANGER: H: 225, 319, 302, 312, 332; P: 210, 280, 305+351+338).*

Necessary equipment and tools: centrifuge, shaker, ultra sonic device, pipettes, pipette tips

**1. Preparation of sample solution**

Dissolve samples (whole contents of tubes 4-10) in each case in 125 µL 0.1% TFA solvent and shake for several seconds.

**2. Preparation of matrix solution**

Produce a mix of ACN and 0.1% TFA in a volume ratio 1:2. Solve HCCA, SA or 2,5-DHB in this solvent up to saturation at room temperature. Assist the solution process by help of ultrasonic device. Spin down excess matrix in a centrifuge (5 min at 14000 rpm) and use only the homogeneous transparent phase. Use matrix solutions according our recommendations in the following table.

| Tube | Sample | Matrix |
|---|---|---|
| 4 | Peptide calibration standard | HCCA, 2,5-DHB |
| 5 | Protein calibration standard I | HCCA, SA, 2,5-DHB |
| 6 | Protein calibration standard II | SA, 2,5-DHB |
| 7 | Tryptic digest of bovine serum albumin | HCCA, 2,5-DHB |
| 8 | Angiotensin II | HCCA, 2,5-DHB |
| 9 | ACTH, clip 18-39 | HCCA, 2,5-DHB |
| 10 | Somatostatin 28 | HCCA, 2,5-DHB |

**3. Preparation onto MALDI-TOF-Target**

According to the dried droplet method, mix equal volumes of sample and matrix solution (HCCA, SA or 2,5-DHB). Apply 1 µL onto a standard steel target and dry at room temperature.

For AnchorChip™ target preparation, refer to the AnchorChip Instructions for Use (# 8215344).

11/13/2023
Analyzer pressure after weekend $5.94 \times 10^{-7}$

— shimadzu folder
current Instrument Status

Administration Console : where you make users

11/13/2023 tool
about 40 seconds
to pump down
after target loads

if you undock laser window
just close it & it will redock

---

Target
1 X 48 2.8 mm wells ( 431 RØØ)

$5 \times 10^{-6}$ mbar

**CA Instruments DSC Q20 — Dry Run (Clickthrough Only)**

al: load empty Tzero pans, run a quick 30→50°C ramp, confirm data save · No sample

**PPLIES (minimal)**

2 empty aluminum Tzero pans + lids (one

eezers (the lab tweezers should be on t

stick for saving .001 file

2 encapsulation press needed for thi

**P**

2 purge gas connected; cell purge

Q20 module (rear switch); wai

20 cooler if equipped (otherwi

ment Explorer (or Trios) on

**NS**

ower button, top of modul

ned lid on the SAMPLE

t lid on the REFEREN

l lids slowly

**UGH**

eriment View

d

0.001 mg (pl

C/min to 50°

eline

001 fi

cell

(2)

Calibration?

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Calibration

other missing parts

Cal. mix

−20°C Freezer

BSA

TOF2 Mix

matrixes

(+) to open data

1

$1.07 \times 10^{-6}$ mbar  $5.0 \times 10^{-6}$ mbar Analysis pressure

Laser power 81

MALDI solution → program
Kratos application

Username: MaldiSolutions Admin
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my users
Smspain
bud Ka$124

pulsed extraction

we only have linear to linear

Laser power range 1 – 180

Solid red 3 services in Windows not ready

Acquire → operate to ramp voltages

remember plate ? turning

Make: Bruker

Model:

Data Installed:

Serial Number:

UNR Asset tag:

General Operationing Instructions

General guide to preping solvents for LCMS

Bruker Start Kit for MADLI-TOF-MS instructions

Calibration instructions

*Bruker MALDI-TOFMS Instructions*

**Rules**

1) No sample preparation should occur in CB 103, prepare the sample plate in your lab
2) Do not exceed 200 laser shots/sample. The laser has a fixed number of shots it can do and once we hit it the laser must be replaced.
3) Any sample the gets on the instrument should be cleaned up immediately with isopropanol
4) When you leave the instrument make sure the sample carrier mechanism is in the IN position
5) Sign up for your instrumentation time

**To collect data**

1) Launch flexControl from the desktop
2) After the software is fully loaded look in the lower right hand corner to determine whether the sample tray is IN or OUT.

It should be left in the in position when your done analyzing samples for the day.

3) click the eject/insert button (lower left side of window looks like a solid triangle over a line) to eject the sample carrier if it is IN
4) open the sample compartment and place your sample plate in the compartment
5) go back to the software and press the eject/insert button
6) wait for the insertion process to complete
7) load a data acquisition method by clicking the **Select…** button

LP : linear positive (enhanced sensitivity for positive ions)
LN: linear negative (enhanced sensitivity for negative ions)
RP: reflectron positive (enhanced resolution for positive ions)
RN: reflectron negative (enhance resolution for negative ions)

8) Do not modifiy or save over existing methods
9) using the mouse click on the gray circle were your sample is located, this will move the laser and camera to the sample
10) using the mouse in the camera window click on a spot that you want to shoot
11) Click the Start button to begin data collection
12) if you do not see any ions increase the laser power using the slider to the right of the camera window
13) if your baseline is not flat decrease the laser power
14) save your data **File → Save Spectrum to File As…**
15) repeat steps 9 through 14 until all your samples are analyzed
16) eject the sample plate by clicking the eject/insert button
17) open the sample compartment , remove the sample plate, and then ***gently*** close the sample compartment (the latch is made of plastic and will break if you let it free fall)
18) press the sample plate eject/insert button to move the sample carrier into the IN position and then close the flexControl software

**<u>To analyze data</u>**

1) Launch flexAnalysis from the desktop
2) Open your data File → Open Single Analysis…
3) Navigate down your data folder until you get to fid file
4) Click on **Find Mass List** button to label peaks with their corresponding masses
5) Print data by going to **Report → Print…**
6) Close the software when you're done

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## Instructions for Use

# Starter Kit for MALDI-TOF MS

For calibration and testing of matrix-assisted laser desorption and ionization time-of-flight mass spectrometers (MALDI-TOF MS)

### Product description

The kit provides a range of standard peptides and proteins for the purpose of calibration and testing of MALDI-TOF mass spectrometers. This kit is designed to provide standards for most protein and peptide applications. Recrystallized matrices (tube 1-3) with high purity are supplied.

A combination of peptides (tube 4) provides good calibration across a typical molecular mass range between ~1000 and 3500 Da. Two mixtures of proteins (tube 5+6) allow instrument testing and calibrations for the molecular mass range between ~3000 and 70000 Da.

A special mixture of different polypeptide chain fragments (tube 7) was produced by the tryptic digest of bovine serum albumin. Three single peptides (tube 8-10) are provided for adjustment of the mass spectrometer and complete the kit.

All of the standards and matrices in the kit may also be purchased as individual components using the part number listed in the Ordering Information part at the end of this document. The components were investigated and tested on a Bruker Daltonics MALDI-TOF mass spectrometer.

### Kit components

| Tube | Substance | Amount |
|---|---|---|
| | ***Matrices*** | |
| 1 | α-Cyano-4-hydroxycinnamic acid (HCCA) | 200 mg |
| 2 | Sinapinic acid (SA) | 200 mg |
| 3 | 2,5-Dihydroxybenzoic acid (2,5-DHB) | 200 mg |

| Tube | Substance | Amount |
|---|---|---|
| | ***Peptide / protein standards*** | |
| 4 | Peptide calibration standard | for 250 calibration spots |
| 5 | Protein calibration standard I | for 250 calibration spots |
| 6 | Protein standard II | for 250 spots |
| 7 | Tryptic digest of bovine serum albumin | for 250 spots |
| | **Peptides** | |
| 8 | Angiotensin II | 500 pmol |
| 9 | ACTH 18 – 39 | 500 pmol |
| 10 | Somatostatin 28 | 500 pmol |

**Storage and stability**

The kit is shipped at ambient temperatures. We recommend storing all components on arrival at less than 0 °C. Sample solutions should be aliquoted and frozen. We do not recommend refreezing samples after thawing.

**Risk and safety information**

The kit has to be labeled according to Regulation (EC) No 1272/2008. It contains components that are classified as hazardous chemical (Signal words: DANGER and WARNING). Please read the Material Safety Data Sheet which is available for download at www.bruker.com/msds in the product description area.

Beside the kit components, we recommend further chemicals within these Instructions for Use. Please read and observe the respective Material Safety Data Sheet to be provided by your supplier. Observe the general safety regulations when handling chemicals.

**Characterization and measurement results of peptide calibration standard**

The next table contains molecular weights of all components in the peptide mixtures (peptide calibration standard). Fig. 1 shows results of measurement.

| Peptide | $[M+H]^+$ Mono isotopic | $[M+H]^+$ Average |
|---|---|---|
| Angiotensin II | 1046.5418 | 1047.19 |
| Angiotensin I | 1296.6848 | 1297.49 |
| Substance P | 1347.7354 | 1348.64 |
| Bombesin | 1619.8223 | 1620.86 |
| ACTH clip 1-17 | 2093.0862 | 2094.43 |
| ACTH clip 18-39 | 2465.1983 | 2466.68 |
| Somatostatin 28 | 3147.4710 | 3149.57 |

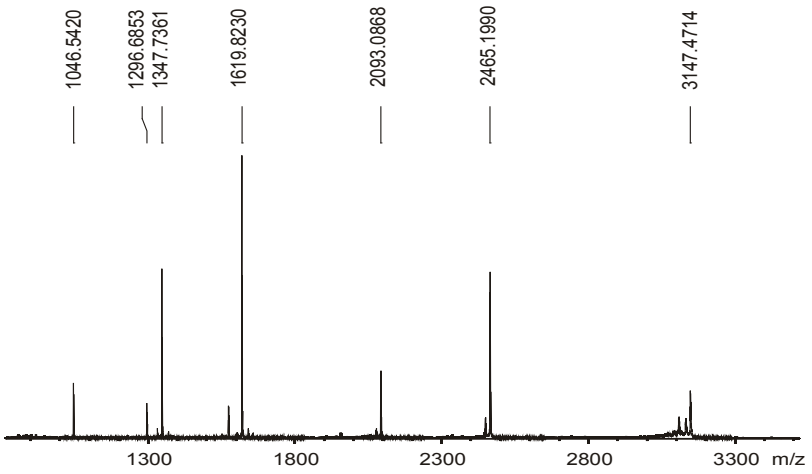


*Fig. 1:* MALDI mass spectrum of peptide calibration standard. Target preparation with HCCA matrix.

**Characterization and measurement results of protein calibration standards**

The kit contains two different protein mixtures (I and II) as test and calibration standards with following compositions.

| Protein standard I | Average m/z | Protein standard II | Average m/z |
|---|---|---|---|
| Insulin $[M+H]^+$ | 5734.51 | Trypsinogen $[M+H]^+$ | 23982 |
| Ubiquitin $[M+H]^+$ | 8565.76 | Protein A $[M+H]^+$ | 44613 |
| Cytochrome C $[M+H]^+$ | 12360.97 | Albumin-Bovine $[M+H]^+$ | 66431* |
| Myoglobin $[M+H]^+$ | 16952.30 | Protein A $[M+2H]^{2+}$ | 22306 |
| Cytochrome C $[M+2H]^{2+}$ | 6180.99 | Albumin-Bovine $[M+2H]^{2+}$ | 33216* |
| Myoglobin $[M+2H]^{2+}$ | 8476.65 | | |

* Bovine serum albumin is not recommended for calibration purposes due to a mass shift to higher m/z values caused by unspecific adduct formation (s. Fig. 4). If an instrument calibration in a higher m/z range is required, we advise an extrapolation of a linear calibration based on the signals at m/z 22306, 23982 and 44613.

The next two figures 2 and 3 show different spectra of protein calibration standard I due to different preparations of MALDI-TOF targets.

Fig. 4 shows the spectrum of protein calibration standard II. The target was preloaded with SA.

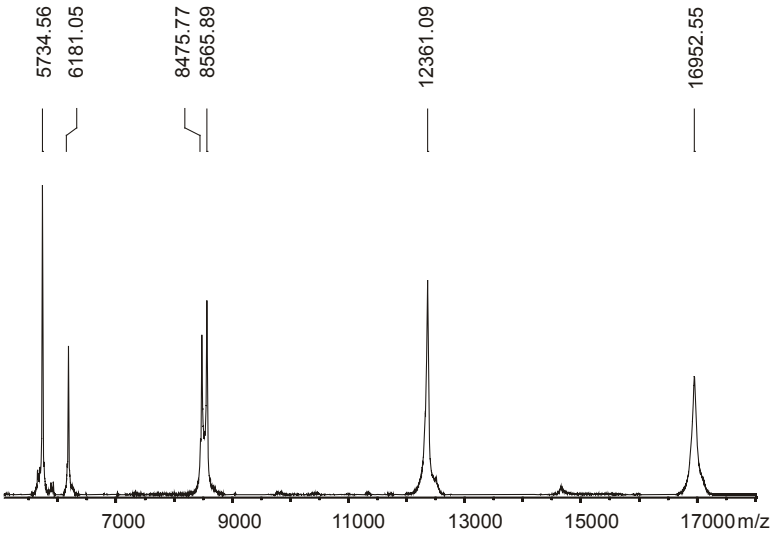


*Fig. 2:* MALDI mass spectrum of protein calibration standard I. Target preparation with HCCA matrix.

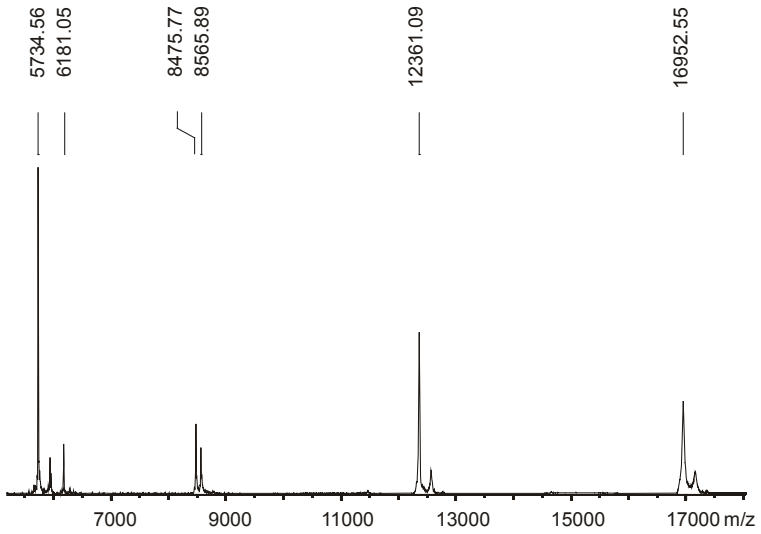


*Fig. 3:* MALDI mass spectrum of protein calibration standard I. Target preparation with SA matrix.

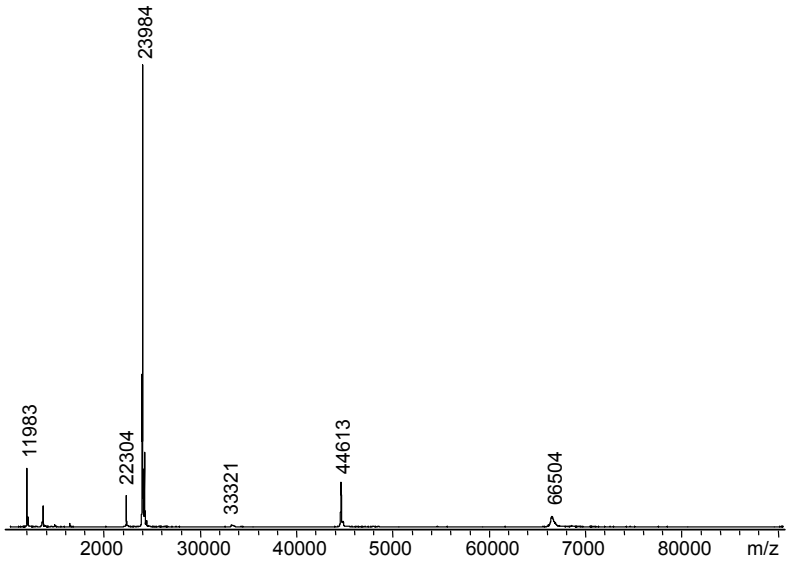


*Fig. 4*: MALDI mass spectrum of protein calibration standard II. Target preparation with SA matrix, linear calibration with protein A (1- and 2-fold charged) and trypsinogen (1-fold charged).

**Characterization and measurement results of tryptic digest of bovine serum albumin**

Different proteolytic polypeptides characterize the tryptic digest of bovine serum albumin (BSA). In the next table the different peptides and molecular masses are listed. Figure 5 depicts a typical MALDI-TOF spectrum of the BSA digest.

| Fragments | $[M+H]^+$ Mono isotopic | $[M+H]^+$ Average |
|---|---|---|
| Intact protein | | 66431.00 |
| [161 – 167] | 927.493 | 928.06 |
| [66 – 75] | 1163.631 | 1164.33 |
| [361 – 371] | 1283.711 | 1284.49 |
| [402 – 412] | 1305.716 | 1306.49 |
| [569 – 580] | 1399.693 | 1400.62 |
| [360 – 371] | 1439.812 | 1440.67 |
| [421 – 433] | 1479.795 | 1480.69 |
| [347 – 359] | 1567.743 | 1568.71 |
| [437 – 451] | 1639.938 | 1640.90 |
| [347 – 360] | 1723.844 | 1724.89 |
| [508 – 523] | 1880.921 | 1882.13 |
| [529 – 544] | 1907.921 | 1909.15 |
| [168 – 183] | 2045.028 | 2046.31 |

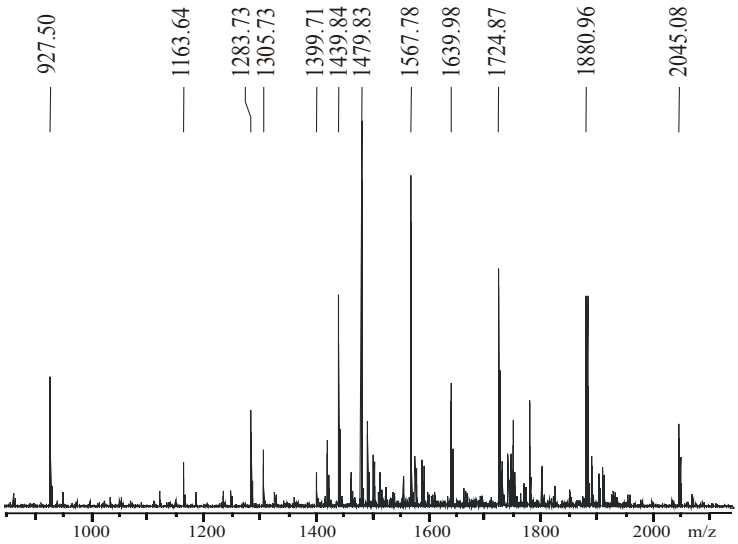


*Fig. 5:* MALDI mass spectrum of tryptic digest of bovine serum albumin. Anchor-Chip™ target preparation with HCCA matrix.

**Sample preparation procedure (recommendation)**

***Pre-Remarks***
Poor sample preparation will degrade sensitivity, yield low resolution and poor reproducibility. The generation of ions through MALDI depends on the production of a suitable composite material, consisting of the matrix substance and the analyte. For best results use only chemicals of highest available purity.

***Chemicals and materials required***
Acetonitrile (ACN)
0.1% Trifluoroacetic acid (0.1% TFA) in water

: *Acetonitrile is classified as hazardous chemical (DANGER: H: 225, 319, 302, 312, 332; P: 210, 280, 305+351+338).*

Necessary equipment and tools: centrifuge, shaker, ultra sonic device, pipettes, pipette tips

**1. Preparation of sample solution**
Dissolve samples (whole contents of tubes 4-10) in each case in 125 µL 0.1% TFA solvent and shake for several seconds.

**2. Preparation of matrix solution**
Produce a mix of ACN and 0.1% TFA in a volume ratio 1:2. Solve HCCA, SA or 2,5-DHB in this solvent up to saturation at room temperature. Assist the solution process by help of ultrasonic device. Spin down excess matrix in a centrifuge (5 min at 14000 rpm) and use only the homogeneous transparent phase. Use matrix solutions according our recommendations in the following table.

| Tube | Sample | Matrix |
|---|---|---|
| 4 | Peptide calibration standard | HCCA, 2,5-DHB |
| 5 | Protein calibration standard I | HCCA, SA, 2,5-DHB |
| 6 | Protein calibration standard II | SA, 2,5-DHB |
| 7 | Tryptic digest of bovine serum albumin | HCCA, 2,5-DHB |
| 8 | Angiotensin II | HCCA, 2,5-DHB |
| 9 | ACTH, clip 18-39 | HCCA, 2,5-DHB |
| 10 | Somatostatin 28 | HCCA, 2,5-DHB |

**3. Preparation onto MALDI-TOF-Target**
According to the dried droplet method, mix equal volumes of sample and matrix solution (HCCA, SA or 2,5-DHB). Apply 1 µL onto a standard steel target and dry at room temperature.

For AnchorChip™ target preparation, refer to the AnchorChip Instructions for Use (# 8215344).

## Ordering Information

| Product | Part No. | Size |
|---|---|---|
| Starter Kit | 8208241 | |
| α-Cyano-4-hydroxycinnamic acid (HCCA) | 8201344 | 5 x 200mg |
| Sinapinic acid (SA) | 8201345 | 5 x 200mg |
| 2,5-Dihydroxybenzoic acid (2,5-DHB) | 8201346 | 5 x 200mg |
| Peptide calibration standard | 8206195 | for 5 x 250 calibration spots |
| Peptide calibration standard II | 8222570 | for 5 x 250 calibration spots |
| Protein calibration standard I | 8206355 | for 5 x 250 calibration spots |
| Protein standard II | 8207234 | for 5 x 250 spots |
| Tryptic digest of bovine serum albumin | 8217498 | for 5 x 250 spots |

## Support

E-mail: maldi.appl.support@bruker.com
Phone: +49 (421) 2205-184
Fax: +49 (421) 2205-104

## Sales Information

Email: care@bruker.com
Phone: +49 (421) 2205-0
Web: www.bruker.com/care

For research use only. Not for use in diagnostic procedures.

# 8208241

Revision B, December 2015
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